

MIA by VIA

Bridging the Whitepaper Vision with Enterprise Technical Architecture.

Prepared for Chairman Elijah John Bowdre.

The Vision Made Real

- **The Core Objective:** Replace fragmented physical identification systems with secure, decentralized infrastructure.
- **The Whitepaper Mandate:** Establish a network governed by G-Codes (Government) and C-Nodes (Citizens).
- **The Technical Delivery:** We have engineered a 5-plane architecture isolating Identity, Service, Privacy, Value, and Trust.
- **The Protection:** Zero balance-sheet liability, Zero PII leaks, and Zero origin-database writes.



Network Topology: The Two Primary Nodes

C-Nodes (Citizen)

Non-custodial digital wallets controlled entirely by residents and enterprises.

They securely hold W3C DIDs, store Soulbound Verifiable Credentials (NFTs), and generate client-side zk-SNARK privacy proofs. The citizen's private keys remain exclusively on their device.

G-Codes (Government)

Unique, immutable smart contract identifiers representing specific municipal departments.

Operating as the authoritative source of truth on the Service Plane, G-Codes (e.g., G-MIA-044) evaluate eligibility, issue Verifiable Credentials, and emit tamper-evident cryptographic receipts.

Pillar 1: IDs (Identification Layer)



W3C DIDs

Decentralized Identifiers provide an interoperable standard, ensuring residents and enterprises own their digital roots without reliance on centralized identity providers.



Soulbound NFTs

Official municipal documents like occupational licenses and permits are issued as ERC-5484 Verifiable Credentials—non-transferable assets anchored directly to the citizen's DID.



zk-SNARK Privacy

Citizens prove eligibility (e.g., age, residency) mathematically without disclosing underlying PII. True privacy through Zero-Knowledge Proofs (Groth16) computed client-side.

Pillar 2: Data & Transparency

The One-Way Wall

The public civic platform is strictly isolated from internal county origin databases. Public verification layers can read status trees but are physically incapable of executing writes to government core systems, eliminating ransomware vulnerabilities.

OPEN TRUST Verification

Every transaction emits an EIP-712 signed receipt. The Attestation Engine blocks unverified claims, enforcing transparency through an immutable Open Checkbook.



Pillar 3: Dollars (Tripartite Custodial Settlement)



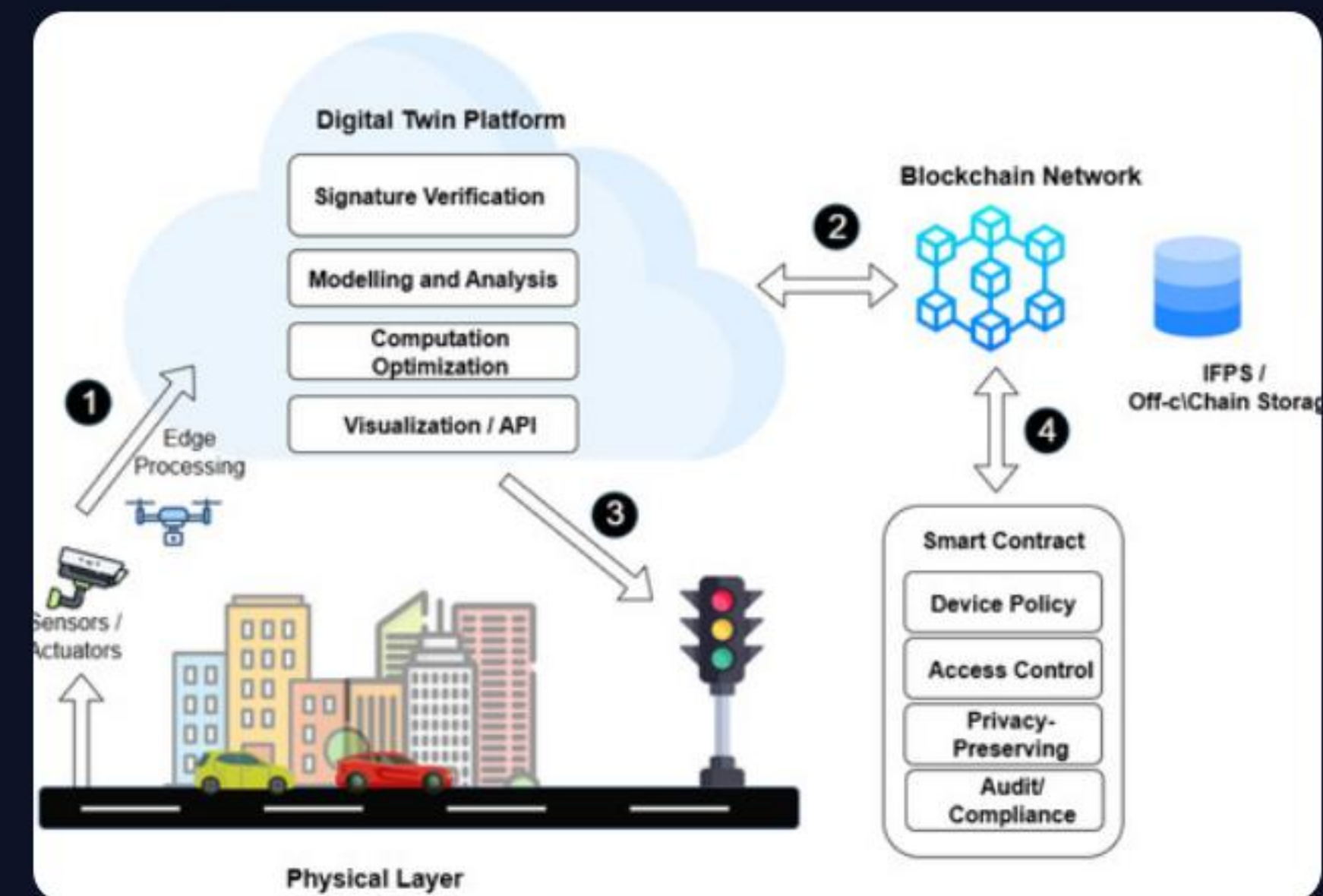
BitGo Bank (Digital Lock)

OCC National Trust. Manages cryptographic private keys, multi-sig cold storage, and EIP-712 compliance.



USTIB (Physical Lock)

USVI Single-State Charter. Governs physical escrow, vaulted titles, direct Fedwire, and bankruptcy-remote statutory trusts.



StateEngine (Orchestration)

Unykorn middleware. Executes the Dual-Oracle Proof-of-Reserve Mint Gate—never holding funds, only enforcing rules.

The Four Core Design Laws

Law 0: No County Dependency

Operates entirely on public verification data without requiring county budget appropriations, software installations, or rigid vendor lock-in.

Law 1: One-Way Wall

Public verifiers and citizen portals never write to origin government databases, ensuring absolute protection of municipal core systems.

Law 2: Summary Law

Summaries, press releases, or marketing claims cannot serve as supporting evidence. Only primary source documents and signed hashes are valid.

Law 3: Verifiability Without Trust

Every published number, credential, or balance publishes alongside its EIP-712 signed cryptographic payload. Truth by math, not assumption.

Deployment Ready

ANVIL Gates G0–G7 Cleared. Base L2 Prepared for Initial Departmental Pilot.

VIRTUAL INFRASTRUCTURE ASSETS | [VIA.MIAMI](https://via.miami)

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